Aryaman Raj Jaiswal

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Education

Netaji Subhas University of Technology Dwarka, Delhi
Bachelor of Technology in Electronics and Communications *Aug. 2024 – Present*

Experience

Bharti Airtel Limited June 2026 – Aug. 2026
Software Engineer Intern *Gurugram, India*

- Re-architected an internal **OpenStack** platform tracking live and historical state across a distributed VM fleet; cut dashboard query latency by **30%** via **database migration**, **composite indexing**, and **cursor-based pagination**.
- Designed a guardrailed natural-language interface over the VM database using local **Ollama** LLMs (**LangChain**), **NeMo Guardrails** input/output rails, and a read-only **text-to-SQL** agent, plus an **AI surge-analysis** layer that flags anomalous provisioning spikes and summarizes root causes.

SmartOut Media (Canada) Jan. 2026 – Present
iOS Developer (Contract) *Remote*

- Built the **SmartOut** iOS app from the ground up in **Swift/SwiftUI** (**50K+ installs**) — an offline-first geospatial platform for Ontario hunting and fishing regulations — over a bundled **SQLite** dataset and a migratable **GRDB** store, instrumented with **GA4** analytics and **Crashlytics**.
- Engineered the map layer with **Google Maps SDK** and **GEOSwift**: **456 regulatory polygons** with MULTIPOLYGON and interior-hole support, distance-based marker clustering, polygon hit-testing, and deep links that restore the exact zone, tab, and regulation row across app launches.

Examination Cell, NSUT May 2025 – Aug. 2025
Software Developer Intern *Delhi, India*

- Architected a **Next.js**/**Flask** automation platform serving **10,000+ students**, cutting manual data entry by **70%** with **Pandas**-driven **constraint solvers** for invigilator assignment and room allocation.
- Constructed the **Android** version of **NSUTrack** in **Kotlin** (iOS client alongside), centralizing schedules, attendance, and society notices for **1,500+ students**, with an offline-first cache refreshed by **WorkManager** sync and **FCM** push.

Achievements

- **Apple Swift Student Challenge Winner (2026):** Selected among the top global student developers for **Saldo** (**SwiftUI**, **Vision**, **Core ML**, **on-device LLM**), which runs receipt parsing and spend intelligence entirely offline.
- **Winner, MLH Brainwave 2.0 (DTU):** Led Team Aquacult to **1st place over 200+ teams**, sweeping **Best Overall Project**, **Best Use of Gemini**, and **Best UI/UX Runner-Up**.
- **2x Winner, HackonHills (2024, 2025):** Back-to-back **1st-place** finishes among **125+ teams**, building **Aqua-Sense** (AI/IoT) and **HimYatra** (Mobile/Web) end to end.

Projects

Saldo | *Kotlin, Jetpack Compose, Room, Hilt, Navigation 3, Coroutines/Flow, JUnit, R8* 2026

- Conceived as a native **iOS** app in **Swift/SwiftUI** (**Swift Student Challenge** winner); built the native **Android** version from scratch in **Kotlin**/**Jetpack Compose**.
- Rebuilt the **bank-SMS transaction parser** as a staged rule engine, fixing **13 correctness defects**, including quoted account balances misattributed as transaction amounts and **Rs. 0** rows that corrupted every dashboard aggregate.
- Wrote a **41-case golden corpus** across **9 Indian bank/UPI templates** with a regression test per defect, growing the suite from **0 to 57 passing tests**.
- Designed **idempotent SMS ingest** keyed on a content hash with a watermark advancing only after commit, replacing a full-inbox rescan that ran twice per launch on the **main thread**.
- Migrated the app to **AGP 9 / Kotlin 2.3 / compileSdk 37** with **Navigation 3**, **Room**, and **Hilt**; enabled **R8** with keep rules, cutting the release APK from **32 MB to 5.1 MB**.

AquaSense | *Kotlin, Jetpack Compose, CameraX, Room, WorkManager, WebSocket, FastAPI, LangChain, Gemini* Jan. 2026

- Built the **Android** client in **Kotlin**/**Jetpack Compose** on **MVVM + repository**: **Retrofit** REST and a **WebSocket** voice agent, **CameraX** disease capture, and an offline-first **Room** cache refreshed by **WorkManager** sync driving **StateFlow** recomposition.
- Shipped a **zero-cost voice assistant** on **on-device SpeechRecognizer** with streamed **base64 MP3** playback via **ExoPlayer**, keep-alive ping/pong, **10-turn** session memory, and typed error codes mapped to retryable UI states.
- Grounded every **Gemini** claim in **Random Forest** output to eliminate hallucinated toxicity verdicts, behind a **multi-agent LangChain** layer routing between telemetry health, **computer-vision** disease, and marketplace agents.

Technical Skills

Languages: Kotlin, Java, Swift, Python, C/C++, Go, SQL (PostgreSQL), JavaScript, Dart, HTML/CSS
Android: Jetpack Compose, Coroutines/Flow, Room, Hilt, Navigation 3, WorkManager, DataStore, Retrofit, CameraX, JUnit, R8
AI / ML: On-device LLMs, LangChain, Ollama, NeMo Guardrails, RAG, Text-to-SQL, Core ML, Gemini
Other: SwiftUI, Flask, FastAPI, Next.js, React, Firebase (FCM, Auth), Docker, Supabase, Gradle, Git, Android Studio